Arista Access Node Optical Transport Failure Resolution Procedure for IND-ACC-76

Node: IND-ACC-76 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6171

Description

The failure domain of Optical Transport Failure, denoted by alarm code ALARM_6171, on the Arista access node IND-ACC-76 situated at the Indiranagar Exchange, is precipitated by a concatenation of factors inherent to the optical transport layer. This layer, responsible for transmitting data as light signals through fiber optic cables, is susceptible to degradation of SFP/XFP modules, physical fiber cuts, and fluctuations in receive or transmit power. The Arista equipment, leveraging its EOS (Extensible Operating System), integrates a robust set of diagnostic and monitoring tools to pinpoint the root cause of such failures. The critical severity of this alarm underscores the immediate impact on network availability and necessitates a swift and methodical approach to resolution. Understanding the underlying protocol mechanics, such as the OTN (Optical Transport Network) or DWDM (Dense Wavelength Division Multiplexing) technologies, is crucial for effective troubleshooting. The architectural design of the access node, including its role in aggregating traffic from various sources, further complicates the scenario, as each failure can have a ripple effect on the entire network topology.

Pre-requisite Checks

- Verify remote hands are dispatched to the Indiranagar Exchange
- Confirm the status of redundant links to ensure no single point of failure
- Check for any recent hardware or software changes on the node
- Ensure all necessary spares (e.g., SFP/XFP modules, fiber optic cables) are available

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cuts	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interfaces transceiver, Expected Output: Detailed information about the optical transceivers, including their status and diagnostic data, Description: This command isolates issues related to the optical transceivers, providing insights into potential SFP/XFP degradation or power loss. By examining the output, one can determine if the transceiver is functioning correctly or if it needs replacement.
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Command: show interfaces description, Expected Output: A list of interfaces with their descriptions, including any error messages or alarms, Description: This command helps in identifying any interface-level issues, including those that might be indicative of fiber cuts or other physical layer problems. The descriptions can point to specific interfaces that are experiencing errors or are down.

Command: show running-config interface, Expected Output: The current running configuration for the specified interface, useful for checking settings that might affect optical transport, Description: By examining the running configuration of the affected interface, one can verify if there are any configuration issues or mismatches that could be contributing to the optical transport failure.
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Resolution Steps

Step 1: Step: Replace Faulty SFP/XFP Modules, Command: replace module , Description: If diagnostic commands indicate SFP/XFP degradation, replacing the module with a spare is necessary. Ensure the replacement module is compatible with the Arista equipment and the specific interface.

Step 2: Step: Repair or Replace Fiber Optic Cables, Command: No specific command; involves physical inspection and repair, Description: For fiber cuts or suspected physical damage, dispatching remote hands to inspect and repair or replace the fiber optic cables is essential. This step requires careful handling to avoid further damage.

Step 3: Step: Adjust Rx/Tx Power Settings, Command: interface , tx-power , Description: If power loss is identified as the issue, adjusting the transmit power settings on the affected interface may resolve the problem. This requires careful consideration to avoid over or under powering, which could lead to signal degradation or non-compliance with optical standards.

Post-Check Verification

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Command: show interfaces status, Expected Output: A list of interfaces with their current status, including any that are down or experiencing errors, Description: After completing the resolution steps, this command verifies if the interfaces are now operational and if the optical transport failure has been resolved.

Command: show logging, Expected Output: System log messages, including any error messages related to the optical transport failure, Description: Examining the system logs can confirm if the error messages associated with the alarm code ALARM_6171 have cleared, indicating a successful resolution.
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Roll-back Procedures

- Step: Revert Configuration Changes, Command: rollback configuration last , Description: If the resolution steps introduce new issues or fail to resolve the problem, reverting to the previous configuration may be necessary. This step requires caution to avoid losing any critical configuration changes made since the last known good state.

- Step: Restore Original SFP/XFP Modules, Command: No specific command; involves physical replacement, Description: If replacing the SFP/XFP modules did not resolve the issue or introduced new problems, restoring the original modules (if they were removed) could be a rollback step. Ensure proper handling and testing of the modules before and after reinstallation.